

Notes: Banerjee (QJE, 1992)

A Simple Model of Herd Behavior

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1 Big picture

- **Question.** Can rational people who learn from the choices of earlier decision makers end up herding, and can that herd behavior be socially inefficient?
- **Setting.** The paper studies a sequential choice problem. Agents choose one option from a large set, one after another. Some agents receive private signals about which option is correct, while later agents also observe the choices of earlier agents.
- **Core challenge.** Observing others is individually useful because their actions may reveal information. But once people start copying earlier choices, actions become less informative about private signals.
- **Main theoretical result.** In Bayesian Nash equilibrium, agents often abandon their own private information and follow predecessors. Herds can start very early in the sequence.
- **Main mechanism.** Banerjee’s central idea is the *herd externality*: when one person ignores her signal and follows the crowd, she also destroys information that would have helped later people.
- **Main welfare result.** The equilibrium can be ex ante inefficient. For a large enough population, society can be better off if some early decision makers are prevented from observing others’ choices.
- **Why this matters.** The paper shows that inefficient herd behavior does not require payoff complementarities or agency distortions. It can arise purely from rational social learning with discrete actions.
- **Economic takeaway.** Rational imitation is not necessarily socially beneficial. Information aggregation can fail precisely because individuals rationally use other people’s actions as information.

2 Model setup and key objects

2.1 Agents, options, and payoffs

- There is a population of N agents.
- Agents are risk neutral and maximize expected monetary payoffs.
- There is a continuum of options or “assets” indexed by $i \in [0, 1]$.
- There is a unique correct option i^* such that

$$z(i) = 0 \quad \text{for all } i \neq i^*, \quad z(i^*) = z, \quad z > 0.$$

- Thus, the social problem is a pure discovery problem: if you choose the correct option you get payoff z , and otherwise you get 0.

2.2 Prior beliefs and private signals

- Agents start with a **uniform prior** over which option is correct, so there is no ex ante favored candidate for i^* .

- Each agent receives a signal with probability α .
- Conditional on receiving a signal, it is correct with probability β .
- If the signal is false, it is uniformly distributed on $[0, 1]$, so it is completely uninformative about the true i^* .
- If an agent does not receive a signal, she is completely uninformed.

Interpretation. The model separates two sources of ignorance: some people know nothing, and some people have noisy information. Herding arises because later agents cannot tell which earlier choices came from true signals, false signals, or no signal at all.

2.3 Information structure and timing

- Decisions are made **sequentially**.
- The first agent moves first, the second observes the first agent's choice and then moves, the third observes the first two choices and then moves, and so on.
- Agents observe **past choices**, but not whether predecessors actually had signals, and not what those signals were.
- After everyone has chosen, the alternatives that were chosen are “tested.” Anyone who chose the correct option gets z .

Interpretation. A choice carries noisy social information because it may reflect a private signal. But choices are coarser than signals, which is the fundamental source of information loss.

2.4 Strategies and equilibrium concept

- A strategy is a decision rule mapping each possible history of past choices and the agent's own signal realization into a current action.
- The solution concept is a **Bayesian Nash equilibrium**.

2.5 Tie-breaking assumptions

The paper imposes three tie-breaking assumptions that select the least-herding equilibrium among possible equilibria.

- **Assumption A.** If a decision maker has no signal and everybody before her chose $i = 0$, she chooses $i = 0$.
- **Assumption B.** If a decision maker is indifferent between following her own signal and following someone else's choice, she follows her own signal.
- **Assumption C.** If a decision maker is indifferent between following more than one previous decision maker, she follows the one who chose the highest i .

Interpretation. These assumptions are chosen to minimize herding, so if the paper still gets substantial herding under these rules, the force of the mechanism is strong.

2.6 Ex ante welfare criterion

- Welfare is evaluated **ex ante**, before agents know their position in the sequence.
- This matters because an individual's incentive to copy may be privately optimal when she moves, while still worsening overall information aggregation for society.

3 Models and theoretical results (all equations + interpretation)

3.1 Restaurant example: the core intuition

The introduction gives a transparent two-restaurant example.

- There are restaurants A and B .
- The prior says A is better with probability 0.51 and B with probability 0.49.
- People arrive sequentially and observe previous choices.
- Signals are equally informative.

Suppose the first person gets a signal favoring A and chooses A , while almost everyone else actually has signals favoring B . The second person sees the first choice and has her own signal favoring B . Since the two signals are equally informative, they cancel, and the second person rationally follows the prior and also chooses A .

Interpretation. Once the second person copies the first, her action no longer reveals her own signal. The third person then learns nothing new from her choice and also chooses A , and so on. A herd starts even though aggregate private information strongly favors B .

3.2 First agents' decisions in the basic model

- **First decision maker.** If she has a signal, she follows it. If she has no signal, she chooses $i = 0$ by Assumption A.
- **Second decision maker.**
 - If she has no signal, she imitates the first agent.
 - If she has a signal and the first agent chose a nonzero option, she is indifferent between her own signal and the first agent's signal, so by Assumption B she follows her own signal.
- **Third decision maker.** She may already face a situation in which following others dominates using her own signal.

Interpretation. The first two agents are still relatively informative. From the third agent on, the information conveyed by previous choices may become too compressed, and cascades can begin.

3.3 Lemma 1: when the first two choose the same nonzero option

Lemma 1. If the first and second decision makers have both chosen the same $i \neq 0$, the third decision maker should follow them.

Intuition.

- The first person's nonzero choice reveals that she had a signal.
- The second person's matching action is additional evidence in favor of that same option.
- So the third person rationally follows the repeated option, even if her own signal points elsewhere.

This is the first clean cascade result in the model.

3.4 Proposition 1: the equilibrium decision rule

Under Assumptions A, B, and C, the paper derives a unique equilibrium decision rule D .

1. The first decision maker follows her signal if she has one and chooses $i = 0$ otherwise.
2. For $k > 1$, if the k th decision maker has a signal, she follows her own signal if either:
 - her signal matches some option that has already been chosen, or
 - no option other than $i = 0$ has yet been chosen by more than one person.
3. If some option other than the highest previously chosen i has already been chosen by more than one person, then the informed k th agent chooses that repeated option, unless her signal matches a different previously chosen option.
4. If the option with the highest previously chosen i has been chosen by more than one person and no other nonzero option has been chosen by more than one person, then the informed k th agent chooses that option, unless her signal matches another previously chosen option.
5. If the k th decision maker has *no* signal, she chooses $i = 0$ if and only if everyone else has done so. Otherwise she chooses the highest previously chosen nonzero option, unless some other nonzero option has already been chosen by more than one person, in which case she chooses that repeated option.

Interpretation. The equilibrium rule says that as soon as one option gets sufficiently strong visible support, agents stop treating their private information as decisive. Repetition in observed choices becomes the trigger for a cascade.

3.5 Figure I: schematic of the decision problem

Figure I in the paper summarizes the equilibrium rule for the k th decision maker when $k > 2$.

- If the decision maker has no signal, she either sticks with $i = 0$, follows the highest observed option, or follows an option that has already been chosen by more than one person.
- If the decision maker has a signal, she follows her signal only in relatively weak-information histories; otherwise she follows the crowd unless her signal matches a previously chosen option.

Interpretation. The figure makes visually clear that herding is not a knife-edge possibility. It is the dominant pattern in a large set of histories.

3.6 Probability that nobody discovers the correct option

The paper shows that under the herd equilibrium the probability that *no one* in the population ever chooses the correct option is

$$\Pi = [1 - \alpha(1 - \beta)]^{-1}(1 - \alpha)(1 - \beta).$$

- This probability is decreasing in both α and β .
- If β is sufficiently small, the probability can be arbitrarily close to one.

Interpretation. Even with many agents, social learning need not aggregate information well. The economy can settle on a wrong herd and never recover.

3.7 Comparison with independent decisions

If agents choose without observing each other, the information they possess is independent across people.

- With a large enough population, someone will almost surely choose the correct option.
- The fraction of people choosing the correct option will be close to $\alpha\beta$.

Interpretation. The striking point is that observation of others can make society worse at discovering the truth than if everyone simply relied on their own information.

3.8 Welfare comparison and alternative decision rule D^*

Banerjee constructs an alternative play of the game, D^* , designed to preserve information.

1. If an agent has a signal, she follows it unless someone before her has already followed someone else; in that case she follows suit.
2. If an agent has no signal, she picks an option not yet chosen by anyone else, unless someone before her has already followed someone else; in that case she follows suit.

This rule makes early actions much more informative. With a very large population, the correct choice is revealed with arbitrarily high probability.

The paper gives the probability that by the n th person at least two earlier agents have received the true signal as

$$1 - (1 - \alpha\beta)^{n-1} - (n-1)(1 - \alpha\beta)^{n-2}\alpha\beta.$$

Interpretation. Under D^* , once enough early information accumulates, the whole population coordinates on the correct option. This allows Banerjee to show that for large N , the equilibrium herd outcome is ex ante inferior.

3.9 Destroying information can improve welfare

A particularly sharp welfare implication is that society may benefit from *reducing* information available to early movers.

- If the first n agents are prevented from observing predecessors, they rely only on their own signals.
- If at least two of them choose the same option, later agents infer that it must be correct and coordinate on it.
- For a large enough population, this can make the fraction choosing incorrectly arbitrarily small.

Interpretation. The paper’s famous paradox is that destroying information in a limited sense can improve ex ante welfare because it protects the informativeness of early actions.

3.10 Why herd behavior arises: choices are not sufficient statistics

Banerjee contrasts his model with a Normal learning model in which agents observe predecessors’ choices and those choices *are* sufficient statistics for the underlying information.

- In that Normal model, choices converge to the truth almost surely.
- In Banerjee’s model, choices do *not* summarize all of the information agents have.

Interpretation. This failure of invertibility is the key. Because many different signal realizations can induce the same discrete action, later agents lose information. Herd externality appears precisely because actions compress private signals too much.

3.11 Extensions in Section V

Alternative payoff structures.

The baseline model gives the same reward z to everyone who chooses correctly. The paper then studies a variant where the first person choosing the correct option gets reward z_1 and later correct choosers get z_2 , with

$$0 < \frac{z_2}{z_1} < 1.$$

Proposition 2 shows that herding still occurs, though now a threshold $k^* > 1$ may be required before agents ignore their own signals.

Interpretation. Rewards for originality reduce herding, but do not generally eliminate it.

Alternative information structures.

The baseline assumes each agent observes the full history of earlier choices. The paper argues that many results survive if agents observe only the distribution of past choices rather than their exact order.

Endogenous timing.

If agents can choose when to move and waiting is costly, inefficient outcomes can still occur. In particular, even with low waiting costs, some uninformed agents may move before informed ones.

4 Mechanism and logic (what makes the argument work)

4.1 The herd externality

- When an agent joins the crowd, she is not merely reacting to information.
- She is also changing the information available to future agents.
- By hiding her own signal, she makes subsequent actions less informative.

Interpretation. This is the paper's fundamental welfare mechanism. Herding is not bad just because people copy. It is bad because copying destroys information for those who come later.

4.2 Why discrete actions matter

- Signals live in a rich informational space.
- Actions live in a coarser choice space.
- Many different informational states map into the same action.

Interpretation. If agents could choose a richer action that fully encoded their information, then later agents could invert observed behavior and recover the underlying signals. The paper's result relies on the fact that they cannot do so.

4.3 Role of the tie-breaking assumptions

- Assumption B forces agents to use their own signal when exactly indifferent.
- Assumption C prevents arbitrary coordination on lower-index options.
- The paper explicitly notes that these assumptions are selected to *minimize* the extent of herding.

Interpretation. The result is therefore conservative. If one changed tie-breaking in a more herd-friendly way, welfare properties would generally be even worse.

4.4 Why the model differs from complementarity stories

- In complementarity models, agents may want to do what others do because joint action raises payoffs directly.
- In Banerjee's model, herd behavior arises even without those direct payoff gains.
- The mechanism is purely informational.

4.5 Why the model differs from Scharfstein and Stein (1990)

- Scharfstein and Stein emphasize an agency problem: agents want to persuade a principal that they are right.
- Banerjee shows that no such incentive distortion is needed.

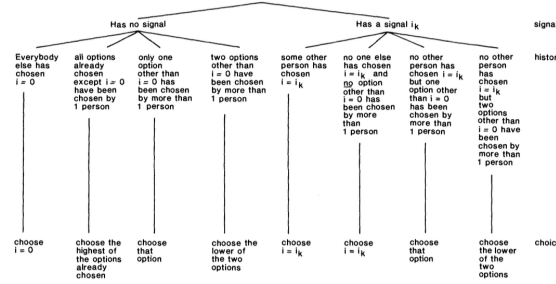


FIGURE I
The k th decision maker's choice problem ($k > 2$)

- In his model, agents capture the payoffs from their own decisions directly, and inefficient herding still arises.

4.6 Limitations the paper acknowledges

- Signals are taken as free; in reality, acquiring information may be costly.
- Options are modeled as a continuum with a unique correct option.
- Agents cannot trade in signals.
- The model abstracts from the possibility that options close to the truth may still be partially valuable.

Interpretation. Banerjee views these as important directions for future research, but argues that the main qualitative mechanism is robust.

5 Tables and figures

5.1 Figure I: the k th decision maker's choice problem

Read:

- Figure I is the paper's only main visual object, and it is very useful.
- It lays out the decision tree for the k th decision maker when $k > 2$.
- The figure makes clear that the equilibrium depends on whether the agent has a signal, whether anyone has already chosen a nonzero option, and whether any option has been chosen by more than one person.
- The main message is visual: once an option obtains repeated support, later choices collapse toward a cascade.

5.2 No empirical tables: what to focus on instead

Read:

- This is a purely theoretical paper, so there are no empirical tables.

- The central quantitative objects are instead the closed-form probability of failure

$$\Pi = [1 - \alpha(1 - \beta)]^{-1}(1 - \alpha)(1 - \beta),$$

and the probability expression used in the welfare comparison under decision rule D^* .

- These formulas matter because they show that the equilibrium can remain highly inefficient even in large populations.

5.3 Restaurant example as a “figure without a figure”

Read:

- The introduction’s restaurant example plays the role that a motivating diagram or empirical event-study would play in a modern paper.
- It shows in one paragraph why an early mistaken cascade can dominate a mass of contrary private information.
- In teaching the paper, this example is the best way to explain the logic before turning to the formal proposition.

6 Short summary

- **Method.** Banerjee studies a sequential Bayesian decision problem with discrete actions and private signals.
- **Mechanism.** People rationally use others’ actions as information, but by doing so they also reduce the informativeness of their own actions for the future.
- **Result.** The unique equilibrium under conservative tie-breaking assumptions features extensive herd behavior.
- **Welfare implication.** Herding can be ex ante inefficient, and society may gain by preventing some early agents from observing others.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Herd behavior can arise from fully rational Bayesian updating in a sequential environment.
- The equilibrium need not aggregate information well.
- The inefficiency comes from an externality in information revelation, not from irrationality, complementarities, or agency per se.

7.2 Economic intuition

- An early action may be informative.
- A later agent copies that action rather than using her own information.
- Once she copies, her action ceases to reveal her private information.
- This makes the next person's problem more like the copier's problem, which produces a cascade.

Interpretation. A herd forms because each agent privately reasons in the same way from a progressively more information-poor public history.

7.3 Takeaway

This paper is important because it explains one of the most famous informational failures in economics with a minimal model. The main lesson is not simply that people copy others. It is that social learning can collapse when actions are too coarse to transmit the information on which they are based. In those situations, rational private behavior can generate persistent, inefficient collective mistakes.